

Jonathan Ortved Melcher

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EXPERTISE: *Climate teleconnections, machine learning, atmospheric physics, tipping points*

EDUCATION

PhD student in Physics

Niels Bohr Institutet, Københavns Universitet, Denmark

May 2025 - Present

- Thesis: Patterns and Causality in Arctic and Mid-Latitude Climate
- The PhD is a part of the Arctic Push project, employing machine learning and other advanced statistics to find novel teleconnections between the Arctic and mid-latitude with a focus on non-linear causal structures.
- Supervised by: Prof. Jens Hesselbjerg Christensen

MSc. in Computational Physics

Niels Bohr Institutet, Københavns Universitet, Denmark

21st of April 2025

- Thesis: Using Novel Explainable Machine Learning Methods to Study Mean Precipitation in Denmark

BSc. in Physics

Niels Bohr Institutet, Københavns Universitet, Denmark

26th of January 2023

- Thesis: Understanding millennial-scale climate change with simple dynamical systems

Exchanges

Eidgenössische Technische Hochschule Zürich, Switzerland

February – August 2024

- Learned about the climate systems at multiple scales and mathematical foundations of machine learning

Université Paris Saclay, France

July – August 2021

- Studied particle physics specifically neutrino masses

Technische Universiteit Eindhoven, Netherlands

July - August 2020

- Researched ideal magnetohydrodynamics for fusion reactors

RESEARCH EXPERIENCE

Research Assistant – Dansgaard-Oeschger Events

with S. O. Rasmussen & G. Vettoretti, Niels Bohr Institutet,

March 2022 – April 2025

- Examined global impact of climate tipping points during the last ice age using ice core data and simulations
- Applied advanced statistics, machine learning, and high-performance computing to analyze climate dynamics
- Published first-author paper in *Climate of the Past* on novel conceptual model for DO event dynamics

Laboratory Assistant

Niels Bohr Institutet, Københavns Universitet, Denmark,

2023 - 2024

- Participated in tephra search in EastGrip ice core and continuous flow analysis campaigns

Field Research Assistant

EastGRIP Station, Greenlandic Ice Sheet, Kalaallit Nunaat,

July – August 2023

- Supported multidisciplinary research team during one-month ice sheet deployment

PUBLICATIONS

Bipolar volcanic ice-core synchronization of the glacial period | *Quaternary Science Reviews*, 2026

Svensson, A., Vettoretti, G., ... **Melcher, J. O.**, et al.

A novel conceptual model for Dansgaard-Oeschger event dynamics | *Climate of the Past*, 2025

Melcher, J. O., Halkjær, S., Ditlevsen, P., Langen, P. L., Vettoretti, G., & Rasmussen, S. O.

Python i LinAlys og MekRel | *python-intro.nbi.ku.dk*, 2021

Melcher, J. O., Severin, J. B., Hedemark, L. S., Nielsen, B. S., & Rasmussen, S. O.

CONFERENCES

Talk: **Data-Driven Discovery of Non-Linear Weather Regimes driving Regional Precipitation Extremes in Europe** | EGU

Jonathan Ortved Melcher, Jens H. Christensen, Chongyang Zhang, Peter L. Langen, and Shuting Yang 2026

Poster: **U-DEM: High spatiotemporal resolution surface elevations of the Canadian Arctic** | EGU

Ann-Sofie Priergaard Zinck, **Jonathan Ortved Melcher**, and Dorthe Dahl-Jensen 2026

Poster: **Beyond PCA: Machine Learning Approaches to Extract Dynamical Drivers** | *NCKF annual symposium*

Melcher, J. O., Christensen, J. H., Zhang, C., Langen, P. L., & Yang, S. 2025

Poster: **Explainable Machine Learning for Climate Prediction** | *NCKF annual symposium*,

Melcher, J. O., Vang, A., Düsterhus, A., Sørup, H. D., & Christensen, J. H. 2024

Poster: **Using Principal Component Analysis for Comparing Climate Models - A Case Study Comparing LOVECLIM and CCSM** | *Willi Dansgaard Centenary, Copenhagen*

Melcher, J. O., Halkjær, S., Rasmussen, S. O., & Vettoretti, G. 2023

Poster: **Understanding the heterogeneous global climate imprint of Dansgaard-Oeschger Events and Heinrich Events - categorizing the events** | *IPICS, Crans Montana*

Melcher, J. O., Halkjær, S., Rasmussen, S. O., & Vettoretti, G. 2022

TEACHING

Teaching Assistant

Niels Bohr Institutet, Københavns Universitet, Denmark,

September 2020 - Present

- Taught courses across mechanics, climate physics, data science, and mathematics to 500+ students over five years
- Served as Python teaching co-lead for mechanics lab, developing materials and leading programming instruction
- Delivered guest lectures in Climate Physics (2023, 2024), making meteorology accessible to physicists

AWARDS & GRANTS

Grants for semester in Zürich, 85 000 dkk 2023-2024

- Awarded 6 different grants ranging from 5 000 dkk to 30 000 dkk.

Fysisk Studenterforenings Legat, 42 dkk 2023

- An award given by the students to a student. Received for expanding the community in the Revue and for improving working conditions for TAs at the Niels Bohr Institute

Københavns Universitets Studiemiljøpris, 25 000 dkk 2023

- Given for our work in promoting unity across cohorts and disciplines, and our ability to highlight and contextualize the mental stresses experienced by students.

OUTREACH

Public lecture “Strøbys Grønlandsfarer” 15th of April 2026

Strøby Kirke, Strøby

- An invited talk on field experience and climate change consequences, [media appearance](#)

Dorm library talk “slåbrok” 10th of June 2025

Regensen, Copenhagen

- An invited talk explaining how we know the climate is changing

High school class inspirational talk 22nd of November 2023

Rysensteen Gymnasium, Copenhagen

- An invited talk on ice core science, tipping points and life on a polar research station

Science communicator at Roskilde Festival 24-27th of July 2023

Science Pavilion, Roskilde Festival

- Explained climate change mechanisms and solutions to festival attendees

LEADERSHIP & ACTIVITIES

Supervision

- Co-supervised a project outside course scope and a master's thesis, using explainable machine learning and causality networks respectively

Trusted Reviewer for Institute of Physics

2025

Stage Manager/Coordinator

October 2021 – June 2023

Physics Revue (Fysikrevyen)

- Led 80+ volunteers producing annual performances for 800+ paying guests with 200,000 kr budget

Union Representative for Teaching Assistants

2022 – 2023

Niels Bohr Institutet, Københavns Universitet, Denmark

- Advocated for improved working conditions focusing on paid hours and management communication